Linear Algebra

K. Höflinger

Linear algebra is the mathematical subject of modules and linear spaces together with the linear mappings defined between them. It is the fundament for a couple of other topics in mathematics, which relies on the concepts provided by linear algebra.

The content is divided up into three main parts, namely *rings and fields*, *linear spaces and mappings* and finally *algebras*. Let us first recall from group theory that an *Abelian group* is a set G together with an associative operation $+ : G \times G \to G$ such that

1. there is exactly one element $0 \in G$ such that the computation rule $x + 0 = 0 + x = x$ holds for all other $x \in G$, and 0 is called the *unit element* in G.
2. to each $x \in G$ there is a unique element $-x \in G$, called the *inverse* of x such that $x + (-x) = -x + x = 0$.
3. the operation $+$ is *commutative* or *Abelian*, i. e. $x + y = y + x$ for all $x, y \in G$.

I. Rings and Fields

The concepts of *ring* and *field* constitute algebraic structures, which form the base for the domains of numbers $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$. Moreover, rings and fields are used to define further structures like *modules*, *vector spaces* and *algebras*.

In the following let $(R, +)$ be an Abelian group and $\circ : R \times R \to R$ be a further associative operation. The triple $(R, +, \circ)$ is called a *ring*, if the operations $+$ and $\circ$ satisfy the following computation rules, denoted as *distributive laws*:

$$\begin{cases} x \circ (a + b) = (x \circ a) + (x \circ b), \\ (a + b) \circ y = (a \circ y) + (b \circ y), \end{cases}$$

for $a, b, x, y \in R$. The pair $(R, \circ)$ especially forms a semigroup, and the operation $\circ$ is usually called the *ring multiplication* in R.

A ring R is called *unitary*, if there is a element $1 \in R$ such that $x \circ 1 = 1 \circ x = x$ for all $x \in R$. Moreover, a ring is called *commutative*, if the multiplication is so, that is, $(R, \circ)$ is a commutativee semigroup.

Latest Version: 11th November 2026.

Email: *khoeflinger@t-online.de*, Url: https://khoefli.github.io/khMath.

The following examples of rings are well-known:

1. $(\mathbb{Z}, +, \cdot)$ with unit element $1 \in \mathbb{Z}$ is a commutative ring.
2. Let M be a set, $(R, +, \circ)$ be a ring and $\mathcal{A}(M, R)$ be the set of mappings from M to R. We introduce addition and multiplication of mappings $f, g \in \mathcal{A}(M, R)$ according to
$$\begin{cases} (f+g)(x) := f(x) + g(x), \\ (f \cdot g)(x) := f(x) \cdot g(x), \end{cases}$$
for all $x \in M$. The set $(\mathcal{A}(M, \mathbb{R}), +, \cdot)$ forms a ring as well.
3. Let $(A, +)$ be an Abelian group and $\mathcal{A}(A)$ be the set of mappings $f : A \to A$. Let for $f, g \in \mathcal{A}(A)$ the sum $f + g$ and the product $f \circ g$ be given by
$$(f+g)(x) := f(x) + g(x), \qquad (f \circ g)(x) := f(g(x))$$
for all $x \in A$. Then $(\mathcal{A}(A), +, \circ)$ is a non-commutative ring with unit element 1_A, where 1_A is the identity mapping in A.
4. The set of (n, n)-matrices with coefficients in $\mathbb{R}$ or $\mathbb{C}$ forms a ring having an unit element with respect to matrix multiplication, which is non-commutative for $n > 1$.

Let us turn to some subclasses of rings by posing some additional requirements regarding the ring multiplication:

Let $(R, +, \circ)$ be a ring. An element $r \in R$ is called a *zero divisor* of R, if $r \neq 0$ and there is $x \in R$ such that $x \neq 0$ and $r \circ x = 0$. A ring R is called *without zero divisors*, if it does not contain a zero divisor.

The absence of a zero divisor can be characterized by means of the *reduction rule*: a ring do not own zero divisors if and only if for $x, y, z \in R$ and $x \neq 0$ the equation $x \circ y = x \circ z$ implies $y = z$. This leads to the notion of *integrity ring*: a commutative unital ring R is called an *integrity ring*, if there are no zero divisors.

Let $(R, +, \circ)$ be a ring with unit element $1 \in R$. An element $r \in R$ is called a *unit* rsp. *invertible*, if there is $r' \in R$ such that $r \circ r' = 1$. If r' exists, then it is uniquely defined by r (notation: $r' = r^{-1}$). If r and r^{-1} are both units, then $(r^{-1})^{-1} = r$ holds.

We denote $E(R)$ to be the set of units of a ring R. The pair $(E(R), \circ)$ forms an Abelian group for every ring R.

1. Subrings. Subrings are defined in a smiliar way as subgroups. Let $(R, +, \circ)$ be a ring. Then a set $U \subset R$ is called a *subring* of R, if for all $r, s \in U$ we have $r + s \in U$ and $r \circ s \in U$.

Hence a subring is closed with respect to the operations $+$ and $\circ$. Each subring $(U,+)$ is an (Abelian) subgroup of $(R,+)$, and $(U,\circ)$ is a sub-semigroup of $(R,\circ)$. The intersection of arbitrarily many subrings of a ring is a subring again. If R is a ring and $S \subset R$ is a non-empty subset, then the *generated subring* $[S]$ of S is defined to be the intersection of all subrings U of R having the property $S \subset U$.

2. Ideals. One of the most important concepts in ring theory is that of *ideal*. Let $(R,+,\circ)$ be a ring and $I \subset R$ be a subring:

1. I is called a *left-sided ideal* in R, if $r \circ x \in I$ for each $r \in R$ and $x \in I$.
2. I is called a *right-sided ideal* in R, if $x \circ r \in I$ for each $r \in R$ and $x \in I$.
3. I is called *two-sided*, if it is a left and right-sided ideal at the same time.

It follows from this definition that for ideals of a *commutative* ring all this notations are equivalent to each other.

The intersection of arbitrarily many ideals in R forms an ideal again. Hence we can assign to each non-empty subset $S \subset R$ a *smallest* ideal containing S. The ideal generated by S is denoted $[S]$ and can be defined as the intersection of all ideals in R containing S.

Let $(R,+,\circ)$ be a ring and $I \neq R$ be an ideal in R.

1. I is called a *maximal ideal*, if it is not contained in a larger ideal besides R itself.
2. I is called a *primary ideal*, if the following is true: If $x, y \in R \setminus I$, then $x \circ y \in R \setminus I$, that is, the set $R \setminus I$ is closed with respect to multiplication.
3. The set $\mathrm{Spec}(R)$ of all primary ideals in R is called the *spectrum* of R. The set of all maximal ideals in R is called the *maximal spectrum* of R.
4. I is called a *principal ideal*, if it is multiplicatively generated by one element, that is, there is a $y \in R$ with $I = \{ x \in R : x = r \cdot y,\ r \in R \}$ rsp. $I = [y]$.

Let R a commutative ring without zero divisors,but with unit element. Then R is called a *principal ideal ring*, if each ideal in R is principal.

1. Each ring R itself is an ideal.

2. The singleton $\{1\}$ with 1 being the unital element of a ring is an ideal.
3. Let I be an interval and $(C(I), +, \cdot)$ be the ring of continuous functions $f : I \to \mathbb{R}$. For $x \in I$ let $I_x := \{f \in C(I) : f(x) = 0\}$. Then I_x is a maximal ideal in $C(I)$. Moreover, each maximal ideal in $C(I)$ is a set I_x for an element $x \in I$.
4. Let $R = (\mathbb{Z}, +, \cdot)$ and $m\mathbb{Z} := \{z \in \mathbb{Z} : z = m \cdot i,\ i \in \mathbb{Z}\}$ for $m \in \mathbb{N}$. Then the sets $m\mathbb{Z}$ are principal ideals in $\mathbb{Z}$. The ring $\mathbb{Z}$ is commutative and does not own zero divisors. Since each ideal in $\mathbb{Z}$ is principal, the ring $\mathbb{Z}$ is a principal ideal ring.

3. Factor Rings. Let $(R, +, \circ)$ be a ring and $I \subset R$ be a two-sided ideal. Since I forms an Abelian subgroup of R, we can consider the (Abelian) factor group $(R/I, +)$.

For each pair of elements $[x], [y] \in R/I$ the multiplicative operation $[x] \circ [y] := [x \circ y]$ is well-defined. The ideal I is two-sided in R, hence the multiplication is independent of the representatives $x \in [x]$ and $y \in [y]$. Now, the triple $(R/I, +, \circ)$ has also the structure of a ring, so the following definition is justified: R/I is called the *factor ring* rsp. the *residual class ring* of R by I.

If R is a commutative ring, then each factor ring R/I is commutative, too, and if R owns a unit element, then for each factor ring R/I a unit element can be constructed, too. For $R = \mathbb{Z}$ and $m \in \mathbb{N}$ the ring $\mathbb{Z}_m := \mathbb{Z}/(m\mathbb{Z})$ is the residual class ring of $\mathbb{Z}$ factorized by the principal ideal $m\mathbb{Z}$. The rings $\mathbb{Z}_m$ constitute commutative rings with unit elements.

Like for other algebraic structures, the notion of *homomorphism* is also defined for rings. Let $(R, +, \circ)$ and $(S, +, \bullet)$ be rings. Then a *ring homomorphism* is a group homomorphism $\phi : R \to S$ that is compatible with the multiplicative operations in R and S in the sense of

$$\phi(x \circ y) = \phi(x) \bullet \phi(y), \qquad (x, y \in R).$$

Other concepts like epimorphisms, monomorphisms, isomorphisms and endomorphisms are defined for rings as well. The rings form together with their homomorphisms a category.

Let $(R, +, \circ)$ be a ring and I be a two-sided ideal in R. Then the residual class assignment $\pi : R \to R/I$, $x \mapsto [x]$ is a ring homomorphism, denoted the *canonical epimorphism* of R to R/I.

The *kernel* $\mathfrak{N}(\phi)$ of a ring homomorphism $\phi: R_1 \to R_2$ is the set of all $x \in R_1$ with $\phi(x) = 0$. If ϕ is surjective, then $\mathfrak{N}(\phi)$ forms a two-sided ideal in R_1. In this case the ring R_1 can be factorized by $\mathfrak{N}(\phi)$, and the following theorem is valid:

THEOREM 3.1. Let R_1, R_2 be rings and $\phi: R_1 \to R_2$ be a surjective ring homomorphism. Then the ideals $R_1/\mathfrak{N}(\phi)$ and R_2 are isomorphic to each other.

In some cases a variant of the concept of ring homomorphisms is meaningful, which is denoted *derivations* over a given ring: Let $(R_1, +, \circ)$ and $(R_2, +, \bullet)$ be rings. Then a *derivative* is a group homomorphism $\phi: R_1 \to R_2$, if it is compatible with the ring multiplications, that is

$$\phi(x \circ y) \;=\; \phi(x) \bullet y \;+\; x \bullet \phi(y), \qquad \text{for all } x, y \in R_1.$$

Look at the following example: let $R := C^1(\mathbb{R})$ be the ring of countinuously differentiable functions $f: \mathbb{R} \to \mathbb{R}$ and $S := C(\mathbb{R})$ be the ring of continuous functions $f: \mathbb{R} \to \mathbb{R}$. Then the differential operator $\phi: R \to S$ defined by $\phi(f) := f'$ for $f \in C^1(\mathbb{R})$ forms a derivation from R to S, since the product rule $(fg)' = f'g + fg'$ holds.

4. Polynomial Rings. For a commutative ring R with unit element let R_0 be the set of *terminating sequences* $f: \mathbb{N} \to R$. It should be clear that for two sequences $p, q \in R_0$ the addition $p + q \in R_0$ is defined component-wise. Moreover, theres is a multiplicative operation $\bullet$ in R_0, which is close to that of convolution:

$$(p \bullet q)_k \;:=\; \Big(\sum_{i+j=k} p_i \cdot q_j\Big), \qquad k \in \mathbb{N}.$$

Hence a ring structure on R_0 is defined by these operations.

The triple $(R_0, +, \bullet)$ is called the *polynomial ring* associated with R, and each element $p \in R_0$ is called a *polynomial.* The unit elements of R_0 are given by the sequences

$$0 := (0, 0, \ldots) \;\text{ and }\; 1 := (1, 0, \ldots).$$

Moreover, if R_0 owns a generating element $x := (0, 1, 0, \ldots)$, i. e. $R_0 = [x]$, then each other element $p \in R_0$ can be represented by the formula

$$p \;=\; a_0 + a_1 \circ x + a_2 + x^2 + \ldots + a_n \circ x^n,$$

which coincides with the usual notations for polynomials. The elements a_i are called the *coefficients* of this polynomial.

By the *degree* of a polynomial p is meant the number $\operatorname{grad} p := \max\{\, a_k \,:\, k \in \mathbb{N}_0 \,\}$. The following computation rules are valid for arbitrary $p, q \in R_0$:

$$\begin{cases} \operatorname{grad} p + q \;\leq\; \max\{\operatorname{grad} p, \operatorname{grad} q\}, \\ \operatorname{grad} p \bullet q \;\leq\; \operatorname{grad} p + \operatorname{grad} q. \end{cases}$$

It is of interest to know that by inserting a ring element $x \in R$ into a polynomal $p \in R_0$ we obtain a ring homomorphism

$$\phi_x : R_0 \to R, \quad p \mapsto p(x),$$

which is often called the *insertion homomorphism.*

5. Fields. By a *field* is meant a ring $(R, +, \circ)$, for which all its elements besides $0 \in R$ own an inverse element. In what follows, let $(K, +, \circ)$ be a ring with unit element 1.

1. K is called a *skew field*, if for each element $x \in K \setminus \{0\}$ there is $x^{-1} \in K$ such that $x \circ x^{-1} = x^{-1} \circ x = 1$.
2. If K is commutative, then K is denoted a *field.*

If K is a skew field, then the pair $(K \setminus \{0\}, \circ)$ forms a group. The sets $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ of rational, real and complex numbers constitute fields, but the set of *quaternions* is only a skew field.

We next answer the question, whether for a given ring R and an ideal I in R the factor ring R/I forms a field. The following result is valid: let $(R, +, \circ)$ be a commutative ring with unit element and I an ideal in R. Then R/I is a field if and only if I is a *maximal* ideal.

Let K be a field and $1 \in K$ be the unit element with respect to multiplication. Then the smallst integer $\chi(K)$ having the property

$$\underbrace{1 + \ldots + 1}_{\chi(K)} = 0$$

is called the *characteristics* of K. If no such number exists, then K is said to have characteristics $\chi(K) = 0$. An example of a field with $\chi(K) = 0$ is given by the field $\mathbb{Q}$ of rational numbers.

For every integrity ring R it is possible to construct a specific field, called the *quotient field* of R. First define an equivalence relation on the cartesian product $M := R \times R \backslash \{0\}$ according to

$$(a,b) \sim (c,d) \;\Leftrightarrow:\; a \circ d = b \circ c.$$

We denote the equivalence class of (a,b) briefly by $\frac{a}{b}$. The equivalence classes of $\sim$ form the basic set Q of the quotient field:

$$Q := M/\sim = \{\frac{a}{b} : (a,b) \in M\}.$$

But addition and multiplication on Q is still required:

$$\frac{a}{b} + \frac{c}{d} := \frac{(a \circ d) + (c \circ b)}{b \circ d}, \qquad \frac{a}{b} \bullet \frac{c}{d} := \frac{a \circ c}{b \circ d}.$$

The reduction rule in integrity rings implies that these operations are independent of the representants of the equivalence classes.

There is still the task to identify the unit elements with respect to addition and multiplication:

$$0 := \frac{0}{1}, \quad 1 := \frac{1}{1}.$$

Then the triple $(Q, +, \bullet)$ forms a field, in which the original ring R can be embedded by an injective mapping ι, that is, there is a monomorphism $\iota : R \to Q$ given by $x \mapsto \frac{x}{1}$.

The most well-known example for this construction is the definition of the field $\mathbb{Q}$ of rational numbers – it is the quotient field of the integrity ring $\mathbb{Z}$.

II. Linear Spaces and Mappings

An Abelian group $(V, +)$ is called a $\mathbb{K}$-*linear space*, if there is an outer multiplication $\cdot : \mathbb{K} \times V \to V$, which fulfills for all $\lambda, \mu \in \mathbb{K}$ and $x, y \in V$

$$\begin{cases} \lambda \cdot (x + y) = \lambda \cdot x + \lambda \cdot y, \\ (\lambda + \mu) \cdot x = \lambda \cdot x + \mu \cdot x, \\ \lambda \cdot (\mu \cdot x) = (\lambda\mu) \cdot x, \\ 1 \cdot x = x. \end{cases}$$

The mapping $(\lambda, x) \mapsto \lambda \cdot x$ (briefly denoted λx) is called *scalar multiplication*. If $\mathbb{K}$ is the field of real or complex numbers, then V is called a *real* or *complex* linear space, respectively. The previous conditions imply $0 \cdot x = 0$ for each $x \in V$.

1. Linear Subspaces. A subset $U \subseteq V$ is called a *subspace* of V, if U is a subgroup of V and $\lambda \cdot x \in U$ holds for all $x \in U, \lambda \in \mathbb{K}$, so U forms a linear space in its own right.

One can verify that the intersection of arbitrarily many subspaces of a given linear space is a subspace again. So for each non-empty set $\mathcal{G} \subset V$ it is possible to define (like in case of groups and rings) a smallest subspace $[\mathcal{G}]$ containing $\mathcal{G}$, which is said to be generated by $\mathcal{G}$: the set $[\mathcal{G}]$ is the intersection of all subspaces $U \supseteq \mathcal{G}$ and forms the smallest subspace in V containing the set $\mathcal{G}$. The subspace $[\mathcal{G}]$ is called the *span* of the set $\mathcal{G}$. Any vector $x \in V$ is an element of $[\mathcal{G}]$ if and only if x can be represented as *linear combination* of vectors in $\mathcal{G}$, that is, there are vectors $x_1, \ldots, x_n \in \mathcal{G}$ and coefficients $\xi_1, \ldots, \xi_n \in \mathbb{K}$ such that $x = \xi_1 x_1 + \xi_2 x_2 + \ldots + \xi_n x_n$.

One of the main concepts in the theory of linear spaces is that of a *Hamel base*. Preliminary, a non-empty set $B \subset V$ is called *linearly independent*, if for each selection $\{e_1, \ldots, e_n\}$ of vectors in B the condition $\xi_1 e_1 + \xi_2 e_2 + \ldots + \xi_n e_n = 0$ implies $\xi_i = 0$ for $i = 1, \ldots, n$. Note that the null vector cannot be an element of any linearly independent set.

A *Hamel base* of a linear space V is a linearly independent subset $\mathfrak{B}$, which generates the entire linear space, i. e. $[\mathfrak{B}] = V$. In this case the set $\mathfrak{B}$ cannot be extended to a larger linearly independent set by adding further vectors $0 \neq v \notin \mathfrak{B}$.

The question arises, whether each linear space has a Hamel basis. Indeed, this is true if the *axiom of choice* in set theory is provided: *assume the axiom of choice to be valid, then each linear space* $V \neq \{0\}$ *has a Hamel basis.* Induced by set inclusion, there is a natural partial order on the family of linearly independent subsets in V, since each chain of linearly independent sets is bounded by the linear space V itself. Due to *Zorn's lemma*, which is equivalent to the axiom of choice, there is a maximal linearly independent set $\mathfrak{B}$ being a Hamel base in V.

Two Hamel bases $\mathfrak{B}, \mathfrak{B}'$ of a linear space V have always identical cardinalities, that is, there is a bijective mapping $\phi : \mathfrak{B} \to \mathfrak{B}'$. By this reason, the linear space V is called *free of rank* $|\mathfrak{B}|$, and we can speak of the *dimension* of a linear space V, which is defined to be the cardinality of any Hamel base in V (we will assign the dimension 0 to the linear space consisting of the null vector only).

If a Hamel base of V consists of a finite number of elements, then the linear space V is called *finite-dimensional* respectively *n-dimensional*, otherwise the space is said to be *infinite-dimensional.*

2. Convexity. A subset $\mathcal{C}$ of a linear space V is called *convex*, if for any selection $v_1, \ldots, v_n$ of elements in $\mathcal{C}$ and coefficients $\lambda_1, \ldots, \lambda_n \in \mathbb{K}$ satisfying $0 \leq \lambda_i \leq 1$ and $\sum_{i=1}^{n} \lambda_i = 1$ the *convex combination*

$$C[v_1, \ldots, v_n; \lambda_1, \ldots, \lambda_n] := \lambda_1 v_1 + \ldots + \lambda_n v_n$$

is again an element of $\mathcal{C}$. The *line segment* $[x, y]$ for $x, y \in V$ is a set of vectors, for which each element $v \in [x, y]$ can be represented by the equation $v = \lambda x + (1 - \lambda) y$ for some $\lambda \in [0, 1]$. In addition, the *inner* line segment (x, y) consists of those vectors, which satisfy this equation for $\lambda \in (0, 1)$. So an alternative definition for the concept of convexity is possible: a subset $\mathcal{C}$ is convex, if and only if $(x, y) \subset \mathcal{C}$ for each pair of vectors $x, y \in \mathcal{C}$.

Notice that for a linear space V the empty set and the space itself are always convex. For example, the convex subsets of the real line are exactly the intervals of finite or infinite type.

Convex sets have the property, that intersections of arbitrarily many convex sets are convex again. This property enables us to declare the *convex hull* $\mathcal{C}(M)$ for each subset M of V, which is defined as the intersection of all convex sets $M' \supseteq M$. The operation, which maps each subset of a linear space to its closed hull, is a hull operator. Thus we have $M \subseteq \mathcal{C}(M)$ and $\mathcal{C}(\mathcal{C}(M)) = \mathcal{C}(M)$.

3. Linear Mappings. Let V and W be arbitrary $\mathbb{K}$-linear spaces. Then each mapping $T : V \to W$ satisfying the condition

$$T(\lambda x + \mu y) \;=\; \lambda T(x) + \mu T(y), \quad (x, y \in V,\ \lambda, \mu \in \mathbb{K})$$

is called *linear transformation* or a *homomorphism* from V to W; often it is called a *linear operator*. We denote $\mathscr{L}(V, W)$ the set of all linear transformations between V and W. Obviously, with the pointwise defined operations $T_1 + T_2$ and λT $(\lambda \in \mathbb{K})$, the set $\mathscr{L}(V, W)$ forms a $\mathbb{K}$-linear space again.

Kernels, Ranges and Cokernels To each linear transformation $T : V \to W$ there are related three further linear spaces, namely the *kernel*, the *range* and the *cokernel* of T. The *kernel* $\mathfrak{N}(T)$ of T is the set of all vectors $x \in V$ satisfying $Tx = 0$, which turns out to be a subspace of V. Similarly, the *range* $\mathfrak{R}(T) := \{ Tx \in W \,:\, x \in V \}$ is a subspace of W.

To define the cokernel of T, we consider the factor group $V/T := V/\mathfrak{N}(T)$. This group is well-defined due to the fact that the subgroup

$\mathfrak{N}(T)$ is *normal* in V. We still have to endow the group V/T with a linear structure: let $\circ : \mathbb{K} \times V/T \to V/T$ be the outer multiplication on V/T defined by

$$\lambda \circ [x] := [\lambda x] \quad \text{for } [x] \in V/T.$$

This operation is independent of the representants in $[x]$, thus V/T is even a $\mathbb{K}$-linear space, called the *cokernel* of V with respect to T.

A vector $x \neq 0$ is called an *eigenvector* to the *eigenvalue* $\lambda \in \mathbb{K}$ with respect to the linear transformation T, if the equation $Tx = \lambda x$ holds. It can be easily shown that the *eigenspace* $E_\lambda(T)$ consisting of all eigenvectors to an eigenvalue λ forms a subspace of V, too.

4. Linear Functionals. As a special case of a linear transformation, a *linear functional* is a mapping $f : V \to \mathbb{K}$ satisfying the condition

$$f(\lambda x + \mu y) = \lambda f(x) + \mu f(y), \quad (x, y \in V,\ \lambda, \mu \in \mathbb{K}).$$

The set $V^* := \mathscr{L}(V, \mathbb{K})$ consisting of all linear functionals on V is clearly a linear space again, called the *algebraic dual space* of V. Notice that if U_1, U_2 are subspaces of V with $U_1 \subset U_2$, then each linear functional $f \in U_2^*$ induces a linear functional $f' \in U_1^*$ by restriction to U_1, that is, we have $U_2^* \subset U_1^*$. This means that it is possible to increase the dual space by decreasing the base space.

Furthermore, the *bidual space* V^{**} of V is the linear space consisting of all linear functionals $\Phi : V^* \to \mathbb{K}$. In particular, each vector $x \in V$ can be sent to an element x^{**} of the bidual space V^{**} by the mapping $\iota : x \mapsto x^{**}(\Phi) := \Phi(x)$ for $\Phi \in V^*$. This mapping is linear, hence its image forms a subspace of V^{**}. It is remarkable that the mapping ι may be defined without any specification of a Hamel base in V. Thus ι is said to be a *natural* linear space homomorphism from V into its bidual space V^{**}.

The embedding $\iota : V \to V^{**}$ is always injective. If the linear space V is finite-dimensional, then ι is actually surjective, and in this case the spaces V and V^{**} are canonically isomorphic to each other. However, in case of an infinite-dimensional linear space no such assertion can be done without further assumptions.

Let V, W be vecor spaces and V^*, W^* their dual spaces. Given a linear transformation $T : V \to W$, each linear functional $f \in W^*$ gives rise to a linear functional $g \in V^*$ by the composition $g := f \circ T$. The assignment $f \mapsto g$ induces a mapping $T^* : W^* \to V^*$, called the

adjoint transformation of T. The adjoint is also linear. Moreover, if T is bijective, then T^* is so.

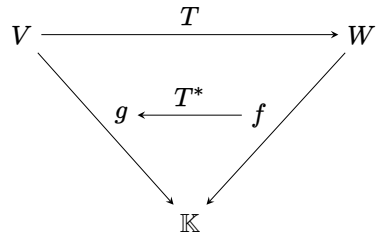


Figure 1: definition of the adjoint transformation.

5. Sesquilinear Forms. Let V be a complex linear space. We denote a *sesquilinear form* on V as a mapping $\mathfrak{a}: V \times V \to \mathbb{C}$, which is linear in the first and conjugate linear in the second argument, that is, for all $x, y, z \in V$ and $\lambda, \mu \in \mathbb{C}$ the following rules hold:

$$\begin{cases} \mathfrak{a}(x+y,z) \;=\; \mathfrak{a}(x,z) \;+\; \mathfrak{a}(y,z)\,, \\ \mathfrak{a}(x,y+z) \;=\; \mathfrak{a}(x,y) \;+\; \mathfrak{a}(x,z)\,, \\ \mathfrak{a}(\lambda x,\mu y) \;=\; \lambda\overline{\mu}\,\mathfrak{a}(x,y)\,. \end{cases}$$

If the underlaying linear space is real, then $\mathfrak{a}$ is also linear in the second argument and thus called a *bilinear form*. The mapping $x \mapsto \mathfrak{a}[x] := \mathfrak{a}(x,x)$ is called the *quadratic form* related to $\mathfrak{a}$.

Furthermore, an *conjugate linear functional* on V is a mapping $f: X \to \mathbb{C}$ having the properties

$$f(x+y) = f(x) + f(y)\,, \quad f(\lambda x) = \overline{\lambda} f(x) \qquad (x, y \in X,\ \lambda \in \mathbb{C}).$$

Then each sesquilinear form $\mathfrak{a}$ on V can be considered as a linear map $A: V \to V^*$, where V^* is the algebraic dual space of V, that is, the linear space of all conjugate linear functionals on V, where addition and scalar multiplication are defined pointwise. Indeed, given a sesquilinear form $\mathfrak{a}$ on V, each vector $x \in V$ gives rise to the conjugate linear functional

$$f_x(y) := \mathfrak{a}(x,y) \quad (y \in V).$$

The mapping $A: V \to V^*$ defined by $A\,x := f_x$ is linear and sometimes called the *representation operator* for $\mathfrak{a}$. Reversely, if $A: V \to V^*$ is linear, then there is a sesquilinear form $\mathfrak{a}: V \times V \to \mathbb{K}$ associated with A according to $\mathfrak{a}(x,y) := (A\,x)y$ for all $x, y \in V$. Hence sesquilinear forms on a complex linear space V are in one-to-one relationship

to linear transformations between the spaces V and V^*.

By an *inner product* in a complex linear space V we mean a specific sesquilinear form $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{C}$ satisfying the additional conditions for all $x, y \in V$:

(I-1) $\langle x, y \rangle = \overline{\langle y, x \rangle}$ *(hermitean symmetry).*
(I-2) $\langle x, x \rangle \geq 0$ *(positivity).*
(I-3) $\langle x, x \rangle = 0$ implies $x = 0$ *(regularity).*

Linear spaces endowed with an inner product $\langle \cdot, \cdot \rangle$ are said to be *unitary spaces* or *inner product spaces.*

Due to the existence of the inner product in V, the concept of *orthogonality* in such spaces is really new in comparison to ordinary linear spaces. Two vectors $x, y \in V$ are called *orthogonal* to each other, if $\langle x, y \rangle = 0$ (notation: $x \perp y$). A subset $\mathfrak{O} \subset V$ is called *orthogonal* or an *orthogonal system*, if any pairs of vectors $x, y \in \mathfrak{O}$ are orthogonal to each other. It is easy to verify that an orthogonal system forms a linearly independent set. For a subset $\mathcal{A} \subset V$ we introduce the *orthogonal space* $\mathcal{A}^\perp$ of $\mathcal{A}$ by means of

$$\mathcal{A}^\perp := \{x \in V : \langle x, y \rangle = 0 \text{ for all } y \in \mathcal{A}\}.$$

III. Algebras

An *algebra* is a linear space $\mathcal{A}$, whose underlying Abelian group has a ring structure being compatible with the scalar multiplication in a certain way. More precisely, let $(\mathcal{A}, +, \circ)$ be a ring and $\mathbb{K}$ be a field, then $\mathcal{A}$ is called a $\mathbb{K}$-*algebra*, if there is an operation $\cdot : \mathbb{K} \times \mathcal{A} \to \mathcal{A}$ such that $(\mathcal{A}, +)$ is a $\mathbb{K}$-linear space satisfying the rule

$$(\lambda \cdot x) \circ y \;=\; x \circ (\lambda \cdot y) = \lambda \cdot (x \circ y) \qquad (\lambda \in \mathbb{K},\, x, y \in \mathcal{A}).$$

An algebra $\mathcal{A}$ is called *unital*, if it has an identity element $1 \in \mathcal{A}$ with respect to the ring operation $\circ$. Finally, $\mathcal{A}$ is called a *commutative* algebra, if $\mathcal{A}$ (considered as a ring) is commutative. For abbreviation one often writes xy instead of $x \circ y$.

1. Algebra Homomorphisms. For some $\mathbb{K}$-algebras $\mathcal{A}$ and $\mathcal{B}$, an *algebra homomorphism* $\Phi : \mathcal{A} \to \mathcal{B}$ is a linear transformation, which is compliant with the ring operation, that is

$$\Phi(x \circ y) \;=\; \Phi(x) \circ \Phi(y) \qquad (x, y \in \mathcal{A}).$$

A bijective character Φ between algebras is called an *algebra isomorphism*, and in this case it can easily be proved that the inverse mapping Φ^{-1} is an algebra homomorphism, too. In the special case, where $\mathcal{B}$ is the field $\mathbb{C}$ of complex numbers, an algebra homomorphism is usually called a *character*. The existence of an algebra homomorphism between arbitrary algebras $\mathcal{A}$ and $\mathcal{B}$ is denoted by $\mathcal{A} \cong \mathcal{B}$.

There is a natural way to endow a non-unital algebra with an identity element. If $\mathcal{A}$ is an $\mathbb{K}$-algebra *without* identity, then an extended algebra *with* identity can be created by $\mathcal{B} := \mathcal{A} \oplus \mathbb{K}$ and $1 := (0, 1)$, where $\mathcal{A} \oplus \mathbb{K}$ is the cartesian product of $\mathcal{A}$ and $\mathbb{K}$ endowed with algebra operations componentwise defined as follows:

$$\begin{cases} (x, \lambda) + (y, \mu) := (x + y, \lambda + \mu) \\ (x, \lambda)(y, \mu) := (xy + \mu x + \lambda y, \lambda\mu) \\ \mu(x, \lambda) := (\mu x, \mu\lambda) \end{cases}$$

for all $x, y \in \mathcal{A}$ and $\lambda, \mu \in \mathbb{K}$. Notice that he original algebra $\mathcal{A}$ is a two-sided ideal within the extended algebra $\mathcal{B}$, and the factor algebra $\mathcal{B}/\mathcal{A}$ is isomorphic to the field $\mathbb{K}$ (considered as a $\mathbb{K}$-algebra): $\mathcal{B}/\mathcal{A} \cong \mathbb{K}$.

2. Subalgebras. Let $(\mathcal{A}, +, \circ)$ be a $\mathbb{K}$-algebra. A subset $\mathcal{U} \subset \mathcal{A}$ is called a *subalgebra* of $\mathcal{A}$, if $\mathcal{U}$ is a subspace of $\mathcal{A}$ and a subring of $(\mathcal{A}, +, \circ)$ together. Similar to subspaces of linear spaces, the intersection of arbitrarily many subalgebras of $\mathcal{A}$ is a subalgebra of $\mathcal{A}$, too. If $\mathcal{G} \subset \mathcal{A}$ is a non-empty subset, then the intersection $[\mathcal{G}]$ of all subalgebras $\mathcal{A}'$ with the property $\mathcal{A}' \supseteq \mathcal{G}$ is called the subalgebra *generated by* $\mathcal{G}$. Finally, if for any subset $\mathcal{G}$ the equality $[\mathcal{G}] = \mathcal{A}$ holds, then $\mathcal{G}$ is called a *generator* of $\mathcal{A}$.

3. Group Algebras. For every group G it is possible to construct an algebra in a natural way by performing the formal linear combinations

$$\alpha_1 \cdot g_1 + \alpha_2 \cdot g_2 + \ldots + \alpha_n \cdot g_n$$

with coefficients $\alpha_i \in \mathbb{R}$. The set $A_{\mathbb{R}}(G)$ of these linear combinations constitutes a $\mathbb{R}$-linear space.

An additional ring structure on $A_{\mathbb{R}}(G)$ is given according to

$$\sum_{i=1}^{n} \alpha_i \cdot g_i \bullet \sum_{j=1}^{m} \beta_j \cdot g_j := \sum_{i,j=1}^{i=n,j=m} \alpha_i \beta_j \cdot g_i g_j.$$

The structure $(A_{\mathbb{R}}(G), +, \bullet)$ is called the *group algebra* of G over $\mathbb{R}$. The construction of the group algebra can be done basically for each ring R.

4. Ideals. A subset $\mathcal{I} \subset \mathcal{A}$ is called an *ideal* in $\mathcal{A}$, if $\mathcal{I}$ is a subspace in the linear space $(\mathcal{A}, +)$ and an ideal in the ring $(\mathcal{A}, +, \circ)$ together. Because the intersection of arbitrarily many ideals in $\mathcal{A}$ is an ideal in $\mathcal{A}$ again, we can assign to each non-empty subset $\mathcal{G} \subset \mathcal{A}$ the minimal ideal $[\mathcal{G}]$, which is called the *ideal generated by* $\mathcal{G}$.

Let $\mathcal{A}/\mathcal{I}$ be the quotient linear space of an ideal $\mathcal{I}$ in $\mathcal{A}$. Then $\mathcal{A}/\mathcal{I}$ is also a subring in $\mathcal{A}$, so it is possible to factorize each algebra $\mathcal{A}$ by an ideal $\mathcal{I}$ in $\mathcal{A}$. The set $\mathcal{A}/\mathcal{I}$ is a $\mathbb{K}$-Algebra again and is called the *factor algebra* of $\mathcal{A}$ by $\mathcal{I}$.